



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



1. Setting up GCP

Set up an account here:

<https://console.cloud.google.com/>

- Use any email account to sign up
- Make sure resource is **No organization**
- Click **New project**

The screenshot shows the Google Cloud console interface. At the top, the 'Google Cloud' logo is visible, followed by a search bar containing 'sctp-dsai-ds2f'. Below the search bar, the 'Welcome' message is displayed, indicating the user is working in the 'sctp-dsai-ds2f' project. The project number is 389866834849 and the project ID is sctp-dsai-ds2f. There are links for 'Dashboard' and 'Cloud Hub'. Below this, there are buttons for 'Create a VM', 'Run a query in BigQuery', 'Deploy an application', and 'Create a storage bucket'. On the right side, there is a 'Join the Google Developer Program' banner. In the bottom right corner, there is a 'New project' button. The 'Select a resource' dropdown menu is open, showing 'No organization' as the selected option. Below the dropdown, there is a search bar for 'Search projects and folders'. At the bottom, there is a table with columns 'Name', 'Type', and 'ID'. The table shows a single entry for 'No organization' with a type of 'Organization' and an ID of '0'.

Google Cloud sctp-dsai-ds2f Search (/) for resources, docs, products, and more Search

Welcome

You're working in sctp-dsai-ds2f

Project number: 389866834849 Project ID: sctp-dsai-ds2f

Dashboard Cloud Hub

Create a VM Run a query in BigQuery Deploy an application

Create a storage bucket

Join the Google Developer Program to learn new skills and dev tools at no cost.

Join today →

Select a resource

No organization ▼

Search projects and folders

Recent Starred All

Name	Type	ID
No organization ?	Organization	0

1. Setting up GCP - Create new Project

- Make sure you are under **No Organization** as shown in the red box

- Choose a name for your GCP Project

Note: GCP project ID has to be globally unique across all GCP users

New Project



You have 10 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[Manage Quotas](#)

Project name *

My Project 26985



Project ID: intricate-shard-490507-d0. It cannot be changed later. [Edit](#)

Organization *

No organization



Select an organization to attach it to a project. This selection can't be changed later.

Parent resource *

No organization

[Browse](#)

Select the organization or folder where this resource will be created

Create

Cancel

1. Setting up GCP - Get Project ID

- You can retrieve your Project ID by clicking the button shown (in the red box)

Select a resource New project

No organization

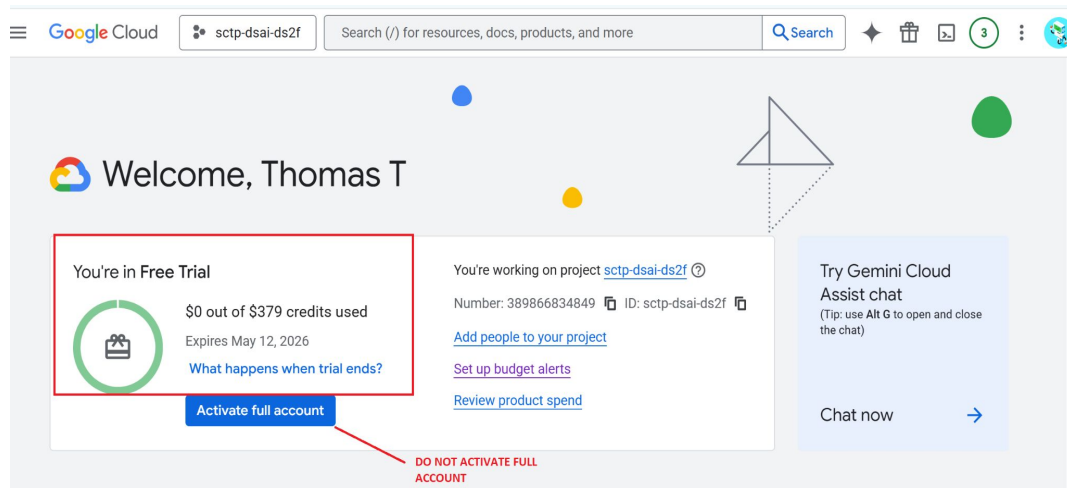
Search projects and folders

Recent Starred All

Name	Type	ID	
My Project 26985 ?	Project	intricate-shard-490507-d0	☆
			☆
No organization ?	Organization	0	
			☆
			☆

1. Setting up GCP - Free Trial

- **Activate free trial**
- **Do NOT activate full account**
- You should see screen similar to the screenshot.



Recommended based on your interest in General ▾

1. Setting up GCP

You might be prompted to create a billing account by entering your credit card details*.

Failure to do so may result in errors in subsequent lessons:

- **Lesson 2.2:** When downloading a file from GCS, you may encounter a "403 error: Billing account not set up."
- **Lesson 2.5:** A similar error may occur when creating tables in BigQuery.
- Use the following link to ensure that your billing account is link to your project: [Verify Billing](#)



Google requires a valid credit card to create a billing account, even when you are on [GCP Free Tier](#). Previous cohorts have successfully used certain prepaid cards available in SG.

2. Install gcloud CLI

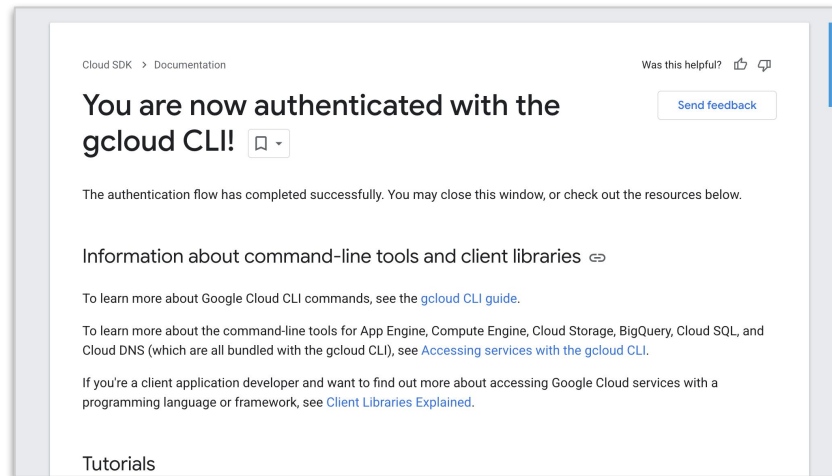
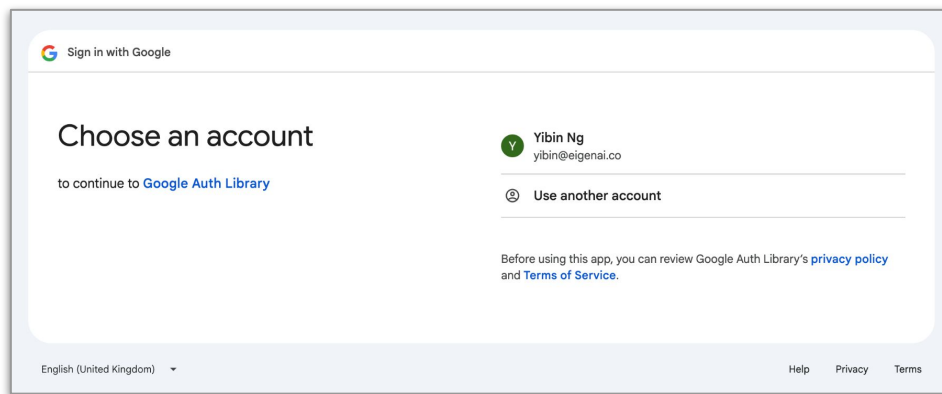
- Mac users, follow instructions here: <https://cloud.google.com/sdk/docs/install>
- Windows WSL users, please follow here: <https://cloud.google.com/sdk/docs/downloads-snap>
- Once `gcloud init` is done, to authenticate your computer to your GCP account, run:
 - WSL users: `gcloud auth application-default login`
 - Mac users: `./google-cloud-sdk/bin/gcloud auth application-default login`
 - Copy and paste the link provided into a browser

```
ngyibin@Ngs-MacBook-Pro Programs % ./google-cloud-sdk/bin/gcloud auth application-default login
Your browser has been opened to visit:

  https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=764086051850-6qr4p6gpi6hn506pt8ejuq83di341hur.apps.googleusercontent.
085%2F&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2
login&state=nSjR3VNjk0bhfCDNATcBe3a1Sws7LP&access_type=offline&code_challenge=XUq26U7TrUC2E-hhA2XwHjZKvpl8xfOTztYP80VxLEq&code_challenge_method=
```

2. Install gcloud CLI

Please log in using the same email account you used to set up your GCP account.



2. Install gcloud CLI

[Optional] Now that you are authenticated, test your setup by typing the commands:

WSL users:

```
bq query --use_legacy_sql=false "SELECT name FROM  
bigquery-public-data.usa_names.usa_1910_current LIMIT 10"
```

MAC users:

```
./google-cloud-sdk/bin/bq query --use_legacy_sql=false "SELECT name  
FROM bigquery-public-data.usa_names.usa_1910_current LIMIT 10"
```

A terminal window with a black background and white text. It displays the output of a SQL query. The output is a table with a single column named 'name'. The table is enclosed in a box with dashed lines at the top and bottom, and vertical lines on the sides. The names listed are Mary, Annie, Anna, Margaret, Helen, Elsie, Lucy, Dorothy, Mary, and Margaret.

name
Mary
Annie
Anna
Margaret
Helen
Elsie
Lucy
Dorothy
Mary
Margaret

You should see the following screen output.

3. Give yourself 'BigQuery Admin' rights

- This is in preparation for the lesson on DBT (Lesson 2.5)
- Search for '**IAM**' in the top search bar
 - **IAM**: Identity and Access Management
 - Click on the 'Edit principal' button next to your email



3. Give yourself 'BigQuery Admin' rights

- Click '+ **Add another role**' - search for '**BigQuery Admin**'
- Click '**Save**'
- You should see this:

IAM

[Allow](#) Deny Recommendations history

Permissions for project "My First Project"

These permissions affect this project and all of its resources. [Learn more](#)

[View by principals](#) View by roles

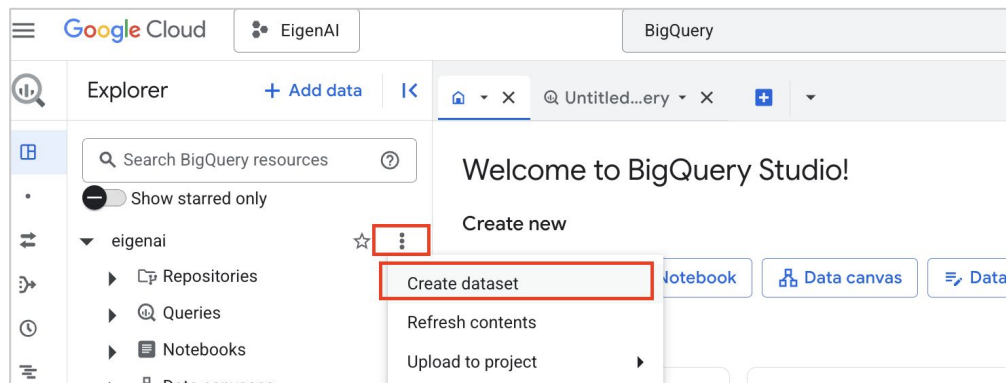
[+ Grant access](#) [- Remove access](#)

[Filter](#) Enter property name or value

☐ Type	Principal ↑	Name	Role	Security insights ?	
☐	wongchenpang@gmail.com	Chen Pang Wong	BigQuery Admin		
			Owner		

4. Create 'snapshots' folder in BQ

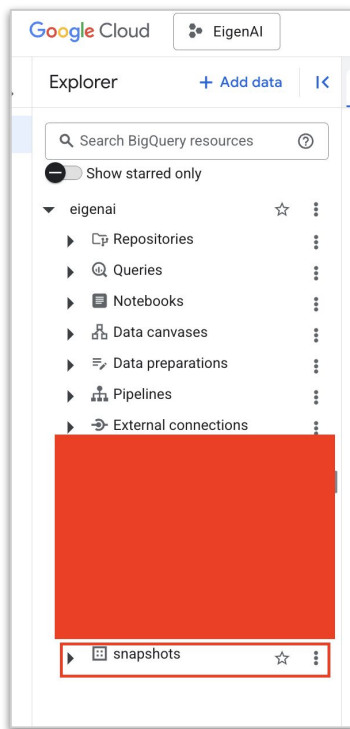
- This is in preparation for the lesson on DBT (Lesson 2.5)
- Go to BigQuery, click on the 3 dots next to your project ID, click '**Create dataset**':



- Enter '**snapshots**' as the Dataset ID
- Use the default location type ('Multi-region' - 'US')
- Click '**Create dataset**'

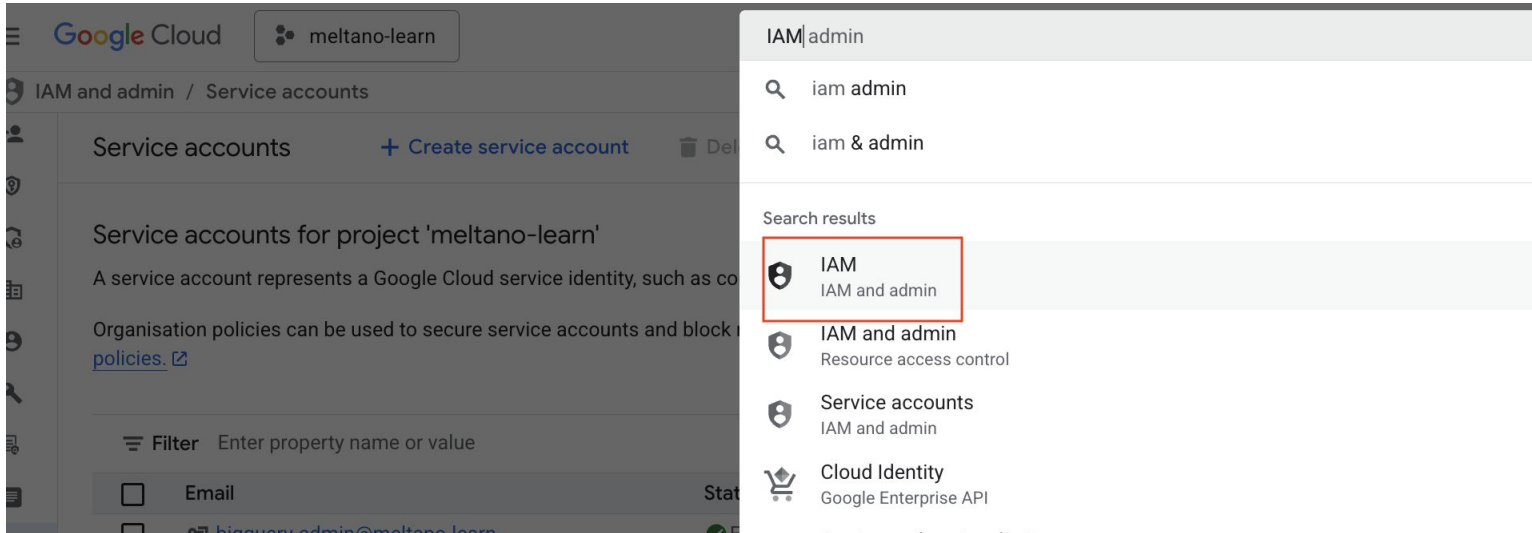
4. Create 'snapshots' folder in BQ

'snapshots' folder created
in your BigQuery:



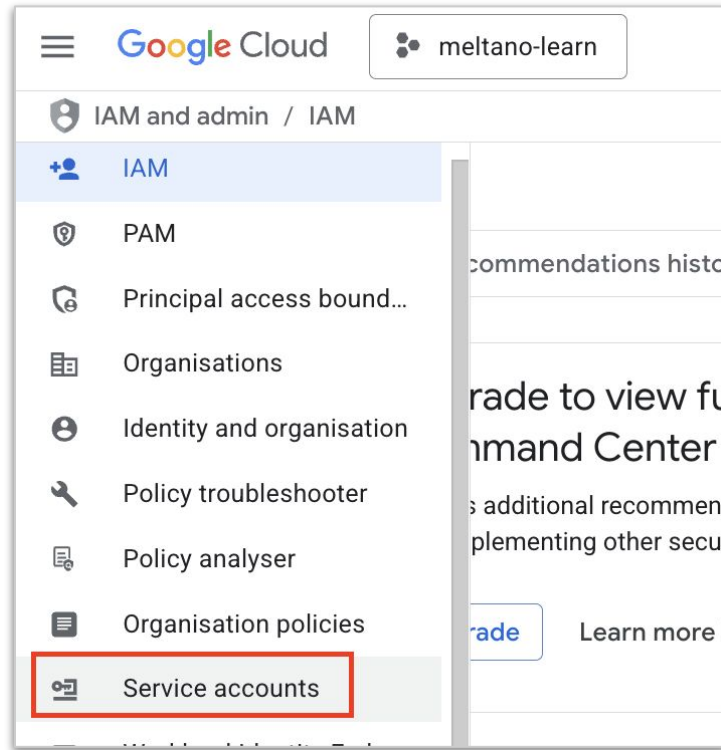
5. Setting up a service account

- This is in preparation for the lesson on Meltano (Lesson 2.6)
- In your Google Cloud project, search for '**IAM**' and click the result:



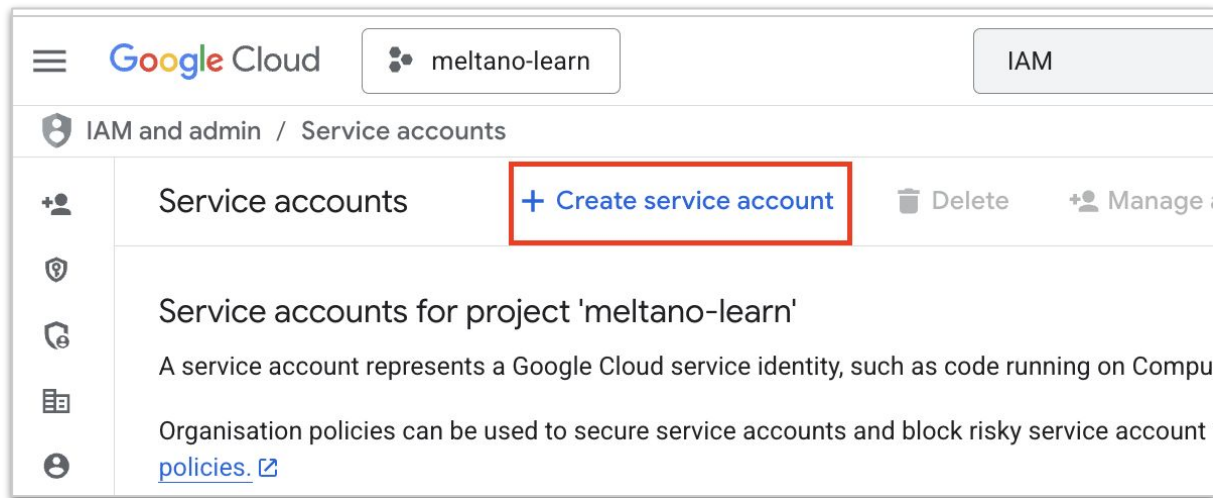
5. Setting up a service account

In IAM, click on '**Service accounts**':



5. Setting up a service account

In IAM, click '**Create service account**':



5. Setting up a service account

Input as shown and click
'Create and continue'

Google Cloud | meltano-learn

IAM and admin / Service accounts / Create service account

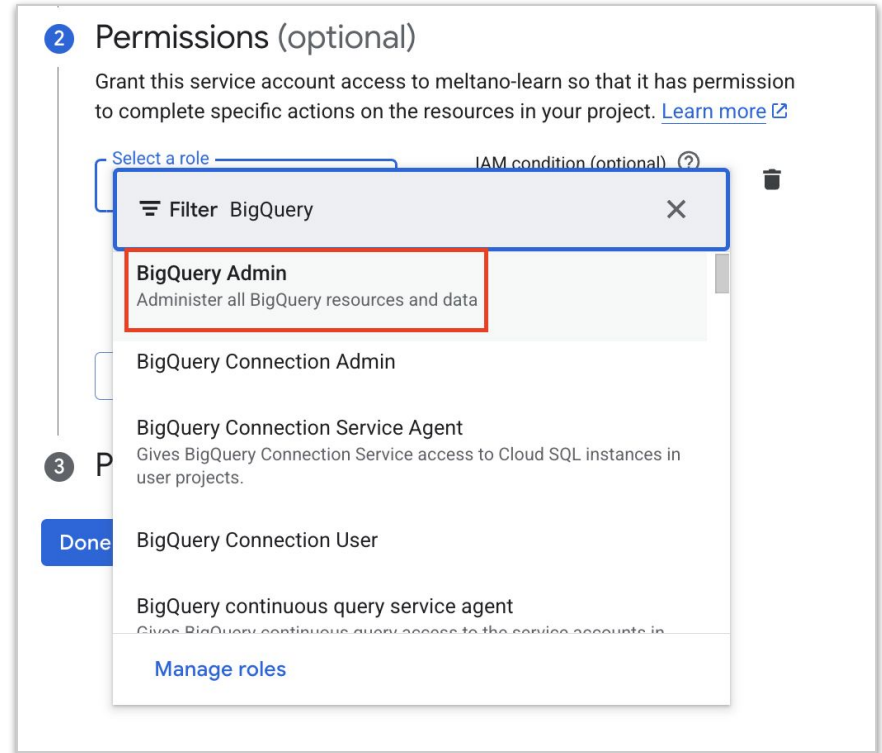
Create service account

- 1 Create service account**
 - Service account name: **bigquery-admin**
Display name for this service account
 - Service account ID *: bigquery-admin-717
 - Email address: bigquery-admin-717@meltano-learn-453807.iam.gserviceaccount.com
 - Service account description: Describe what this service account will do
 - Create and continue**
- 2 Permissions (optional)**
- 3 Principals with access (optional)**

Done Cancel

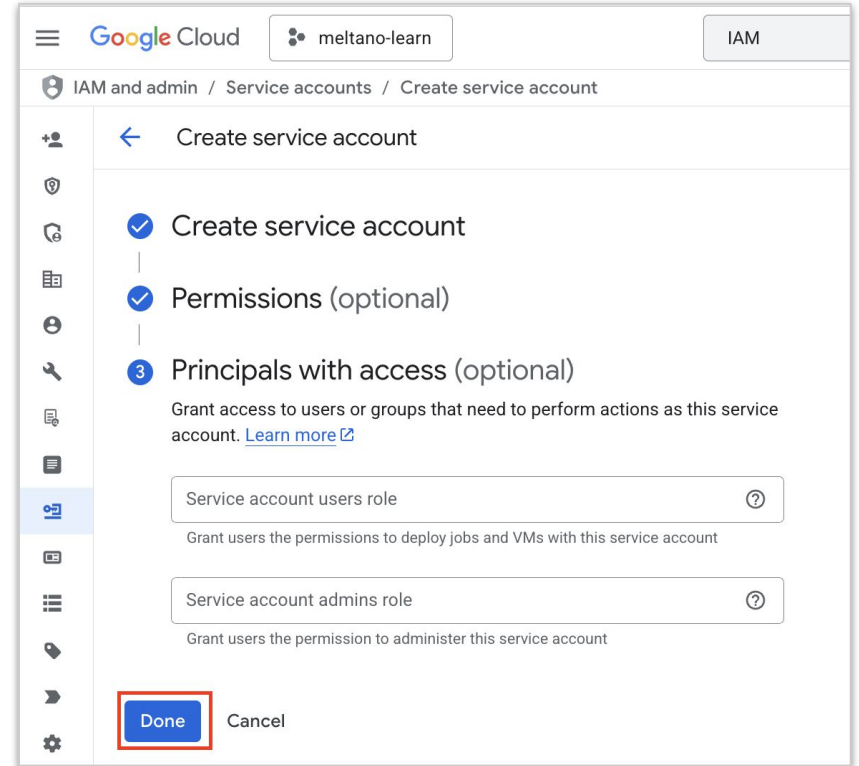
5. Setting up a service account

Under Permissions, search for
'BigQuery Admin' and click
'Continue':



5. Setting up a service account

Skip 'Principals with access'
and click '**Done**':



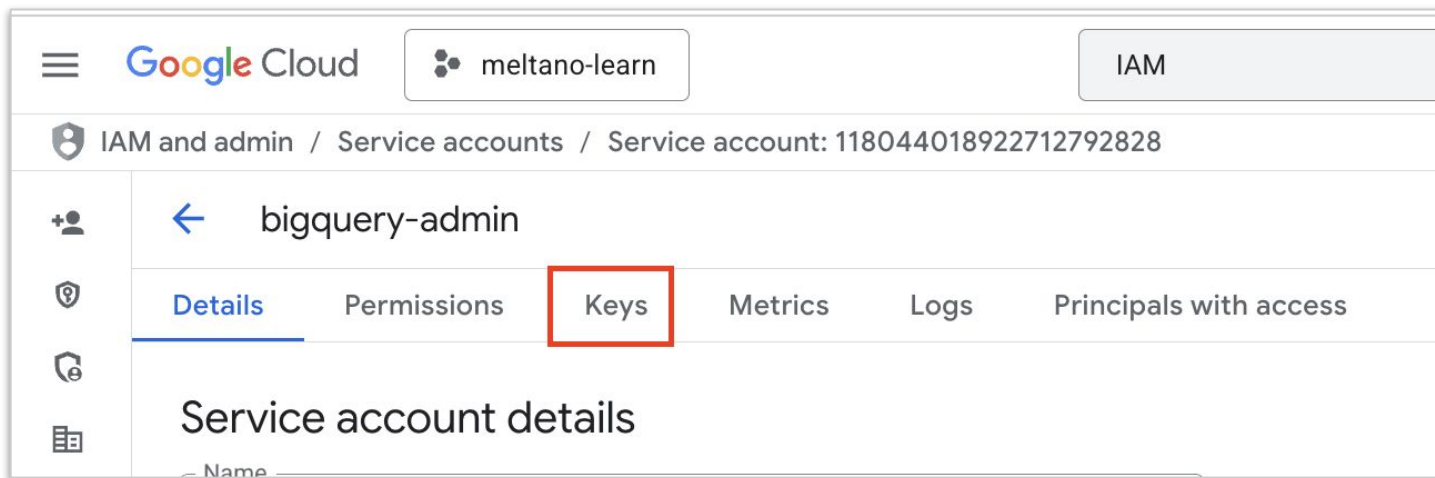
5. Setting up a service account

Click on the service account you just created:

☐	 bigquery-admin-717@meltano-learn-453807.iam.gserviceaccount.com	 Enabled	bigquery-admin
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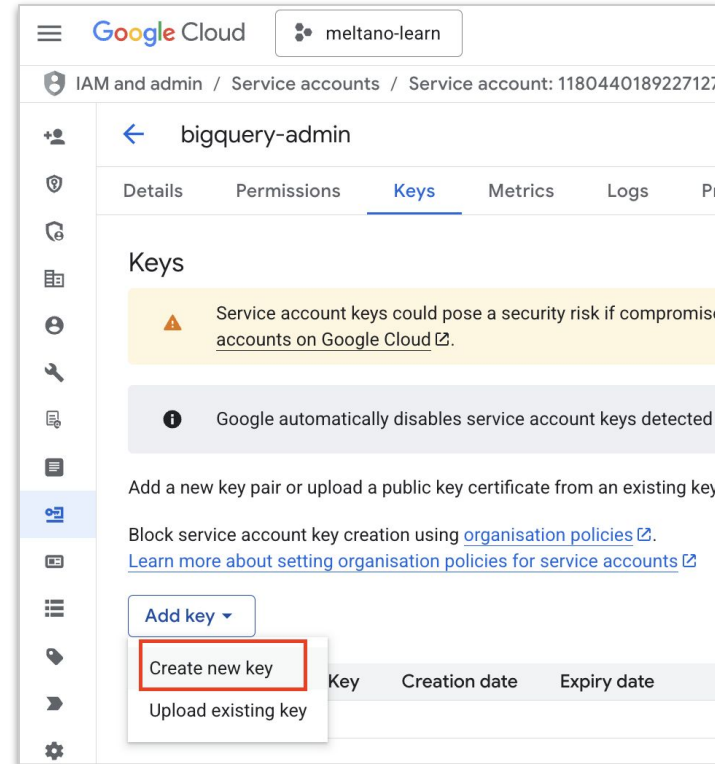
5. Setting up a service account

Click '**Keys**':



5. Setting up a service account

Click '**Create new key**'



5. Setting up a service account

- Select '**JSON**' and click '**Create**'
- A JSON file will be downloaded into your computer
- Put it into your lesson 2.6 folder:

5m-data-2.6-data-pipelines-orchestration/

Create private key for 'bigquery-admin'

Downloads a file that contains the private key. Store the file securely because this key cannot be recovered if lost.

Key type

☒ JSON
Recommended

☐ P12
For backward compatibility with code using the P12 format

Cancel **Create**